

Answer **all** questions in the spaces provided.

Do not write
outside the
box

0 1 . 1 Water has a high heat capacity and a large latent heat of vaporisation.

Describe the importance of each of these properties to living organisms.

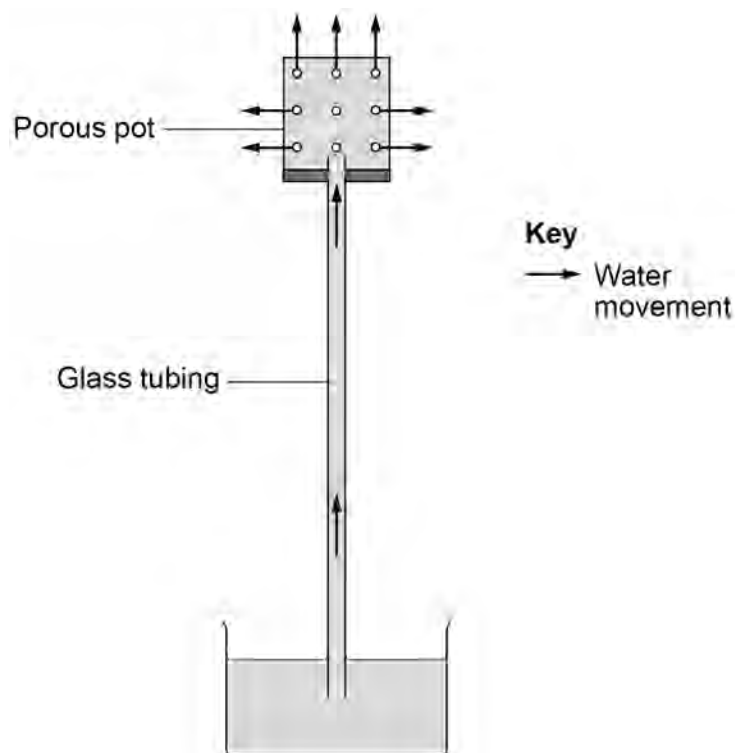
[2 marks]

High heat capacity _____

Large latent heat of vaporisation _____

0 1 . 2 **Figure 1** shows that water loss from a porous pot can cause the upward movement of water.

Figure 1



Biologists have concluded that the experiment in **Figure 1** supports the cohesion–tension theory of water transport in the xylem.

Explain how this conclusion is supported by the experiment in **Figure 1**.

[3 marks]

0 1 . 3

An air bubble was introduced into the glass tubing in **Figure 1**. The air bubble moved a distance (d) of 1.5 cm in 120 minutes. The radius of the lumen (hole) of the glass tubing was 0.6 cm

Use this information and the formula $\pi r^2 d$ to calculate the rate of water movement in the glass tubing in $\text{cm}^3 \text{ hour}^{-1}$.

Use $\pi = 3.14$ in your calculation.

[1 mark]

Answer _____ $\text{cm}^3 \text{ hour}^{-1}$

6

Turn over ►

